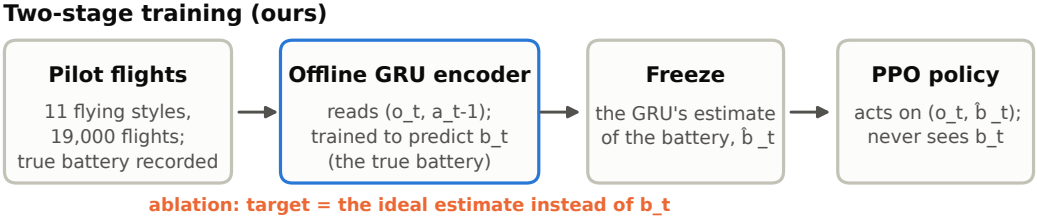
Two-stage training (ours)
Pilot flights
11 flying styles, 19,000 flights; true battery recorded
Offline GRU encoder
reads (o_t, a_t-1); trained to predict b_t (the true battery)
Freeze
the GRU's estimate of the battery, b̂ _t
PPO policy
acts on (o_t, b̂ _t); never sees b_t
ablation: target = the ideal estimate instead of b_t
End-to-end training (fails: the autogenic loop)
Policy flies one narrow style
Battery rarely shows in the flights
Encoder learns a clock instead of the battery
Estimate adds nothing: policy ignores it